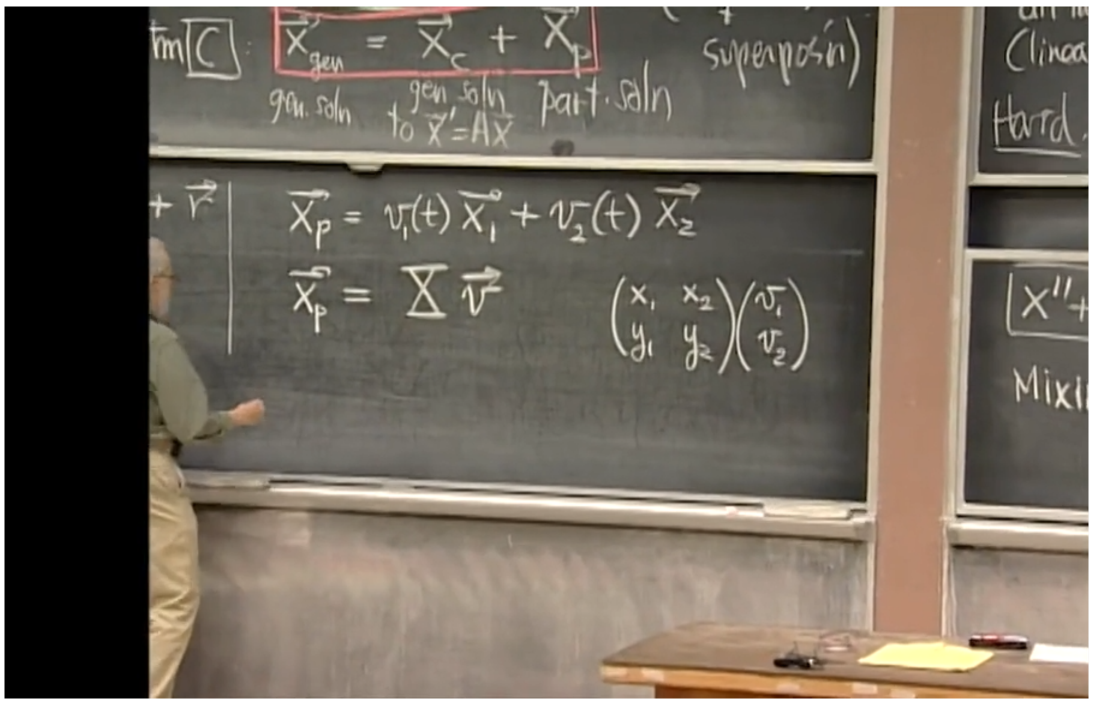
5. Variation of parameters

Variation of parameters



5:59 / 12:58 2.0x

fortunately.

And the same way the bottom thing

will be v1, y1 plus v2, y2.

V1, y1 plus v2, y2.

If I had written it on the other side instead,

which is tempting because the v's occur on the left, here.

That won't work, because what will I get?

I'll get v1, x1 plus v2, y1, which

is not at all what I want.

You must put it on the right.

But this is a very important thing.

It's going to plague us on Monday too.

It must be written on the right and not on the left as a column

vector.

OK?

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Long ago, we used **variation of parameters** to solve single first order inhomogeneous linear ODEs

$$\dot{x} + p(t)\, x = r(t)\,.$$

We first find a solution x_h to the associated homogeneous equation, seek a particular solution of the form $x_p(t) = u(t)\, x_h(t)$, and use the original inhomogeneous ODE to solve for the unknown function $u(t)$.

Now, we are going to use the same idea to solve an inhomogeneous linear $n \times n$ **system** of ODEs:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{r},$$

where $\mathbf{r}$ is a vector-valued function of t.

First, find a basis of solutions to the corresponding homogeneous system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}.$$

Call the basis solutions $\mathbf{x}_1(t), \mathbf{x}_2(t), \cdots, \mathbf{x}_n(t)$. The general homogeneous solution is any linear combination of these:

$$\mathbf{x}_h(t) = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_n\mathbf{x}_n \tag{4.131}$$

$$= \mathbf{X}\mathbf{c} \qquad \text{where } \mathbf{X} = \begin{pmatrix} | & | & & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_n \\ | & | & & | \end{pmatrix}, \ \mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}. \tag{4.132}$$

Notice that in 1 dimension, $x_h(t) = c x_1(t) = x_1(t)\, c$, but in higher dimensions, $\mathbf{x}_h(t) = \mathbf{X}\mathbf{c}$ where $\mathbf{c}$ is a column vector and must be placed to the right of the fundamental matrix $\mathbf{X}$.

Calculator

To find a particular solution, we let the coefficients c_i vary with time. In other words, replace the constant vector $\mathbf{c}$ by the vector **function**

$$\mathbf{v}(t) = \begin{pmatrix} v_1(t) \\ v_2(t) \\ \vdots \\ v_n(t) \end{pmatrix}. \tag{4.133}$$

Now, substitute $\mathbf{x} = \mathbf{X}\mathbf{v}(t)$ in the original system:

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{r} \\ \dot{\mathbf{X}}\mathbf{v} + \mathbf{X}\dot{\mathbf{v}} &= \mathbf{A}\mathbf{X}\mathbf{v} + \mathbf{r} \qquad \text{(product rule of differentiation)} \\ \mathbf{A}\mathbf{X}\mathbf{v} + \mathbf{X}\dot{\mathbf{v}} &= \mathbf{A}\mathbf{X}\mathbf{v} + \mathbf{r} \qquad (\dot{\mathbf{X}} = \mathbf{A}\mathbf{X}) \\ \mathbf{X}\dot{\mathbf{v}} &= \mathbf{r} \\ \dot{\mathbf{v}} &= \mathbf{X}^{-1}\mathbf{r} \qquad (\mathbf{X} \text{ invertible}). \end{aligned}$$

This means

$$\mathbf{v}(t) = \int \mathbf{X}^{-1}\mathbf{r}\, dt. \tag{4.134}$$

and the general solution to the inhomogeneous system is

$$\mathbf{x}(t) = \mathbf{X}\mathbf{v}(t) = \mathbf{X}\left(\int \mathbf{X}^{-1}\mathbf{r}\, dt\right),$$

for any fundamental matrix $\mathbf{X}$ of the associated homogeneous system.

This is a family of solutions because the indefinite integral on the right hand side will result in a constant of integration. Note that the constant of integration is a column vector.

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